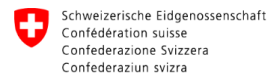


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Safe Use of Explainable AI Methods in Banking

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Agenda

- What is the regulatory scenario?
- What are the problems with XAI methods?
- How can we solve these problems?
- What is the value for banking supervision?



What Is The Regulatory Scenario?



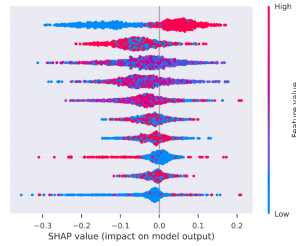
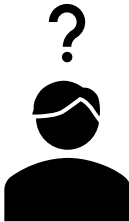
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Bank

We want to use
ML models to
better predict
PD.



one.employee.is_future_manager_1.0
one.employee.costcenter_scaled
one.employee.hourly_rate_scaled
one.employee.date_of_birth_scaled
one.employee.department_scaled
one.employee.gender_Male
one.employee.location_404.0
one.employee.nationality_other
one.employee.exempt_NX
Sum of 5 other features



Supervisor

Ok, but can
you explain the
ML model's
predictions?



How can I trust
that these are
faithful to the
model?



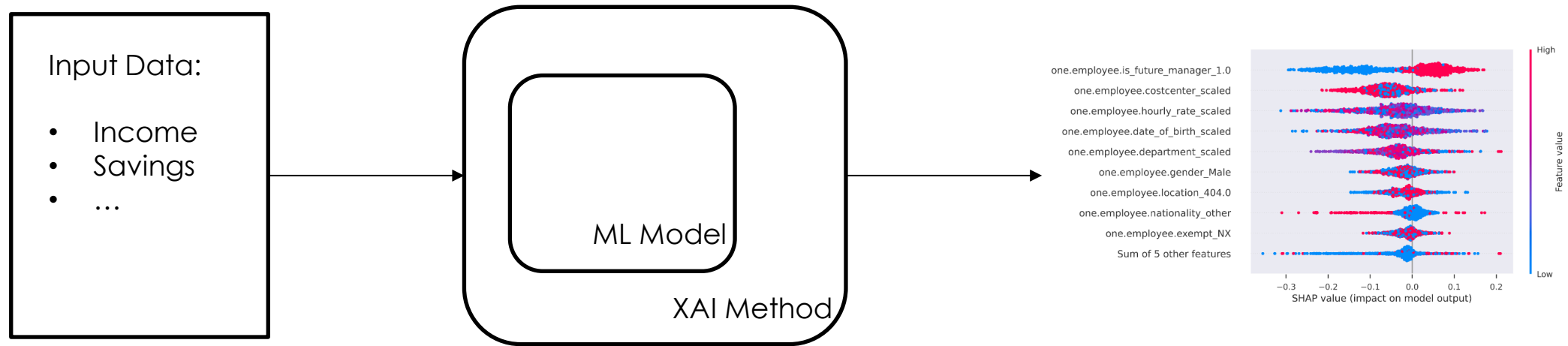
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What Are Problems With XAI Methods?



Problem #1

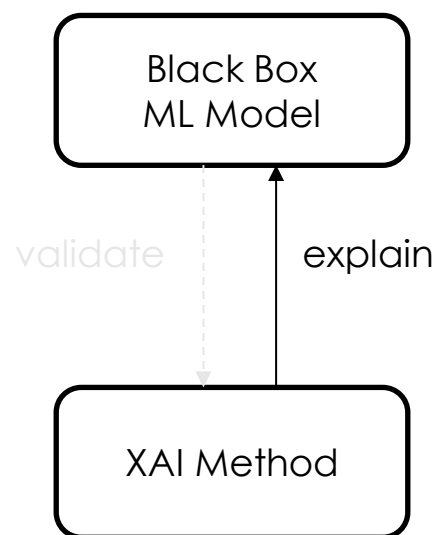
By design, XAI methods always give an explanation.



"Numbers in, numbers out."

Problem #2

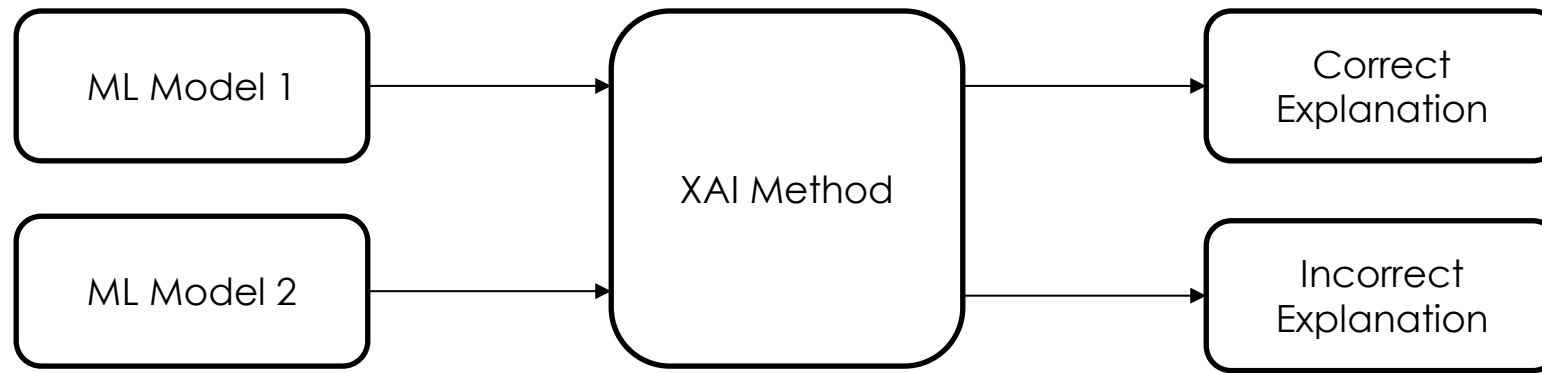
We lack a ground truth to measure the XAI method against.



"We do not know if the model works as the explanations say it does."

Problem #3

We do not know under which conditions we can trust an XAI method.



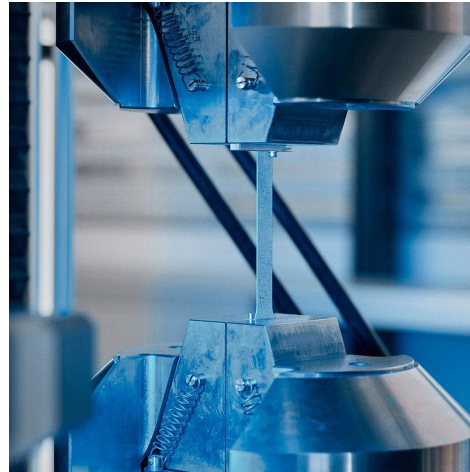
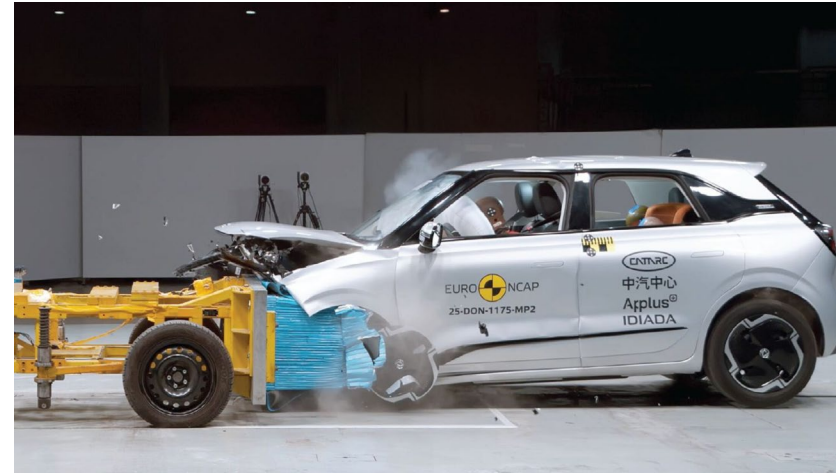
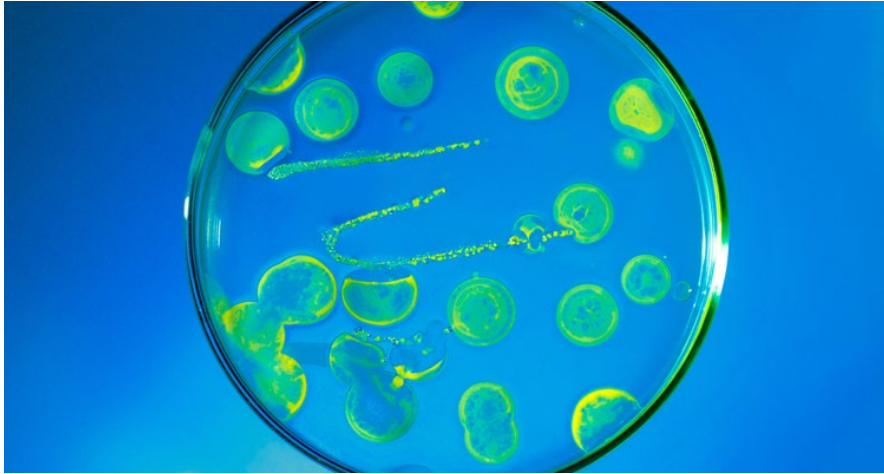
"It depends – but on what?"

How Can We Solve These Problems?



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How where problems like this solved elsewhere?



What we can we learn from them?

1. Defined experiments
2. Known ground truth
3. High degree of control



Research Idea

Create experiments to study XAI method behavior under controlled conditions.

Identify “safe use” conditions for XAI methods.



What is The Value For Banking Supervision?

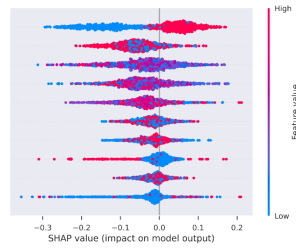


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Bank

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one.employee.location_404.0
one.employee.nationality_other
one.employee.exempt_NX
Sum of 5 other features



Yes/no.

Supervisor

Ok, but can you explain the ML model's predictions?

Is the method used within safe use conditions?



Thank you for your attention! Feedback
or questions?

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